



Bacterial Diarrhea

*(Escherichia coli, Shigella, Campylobacter,
and Salmonella spp)*

SGL 4

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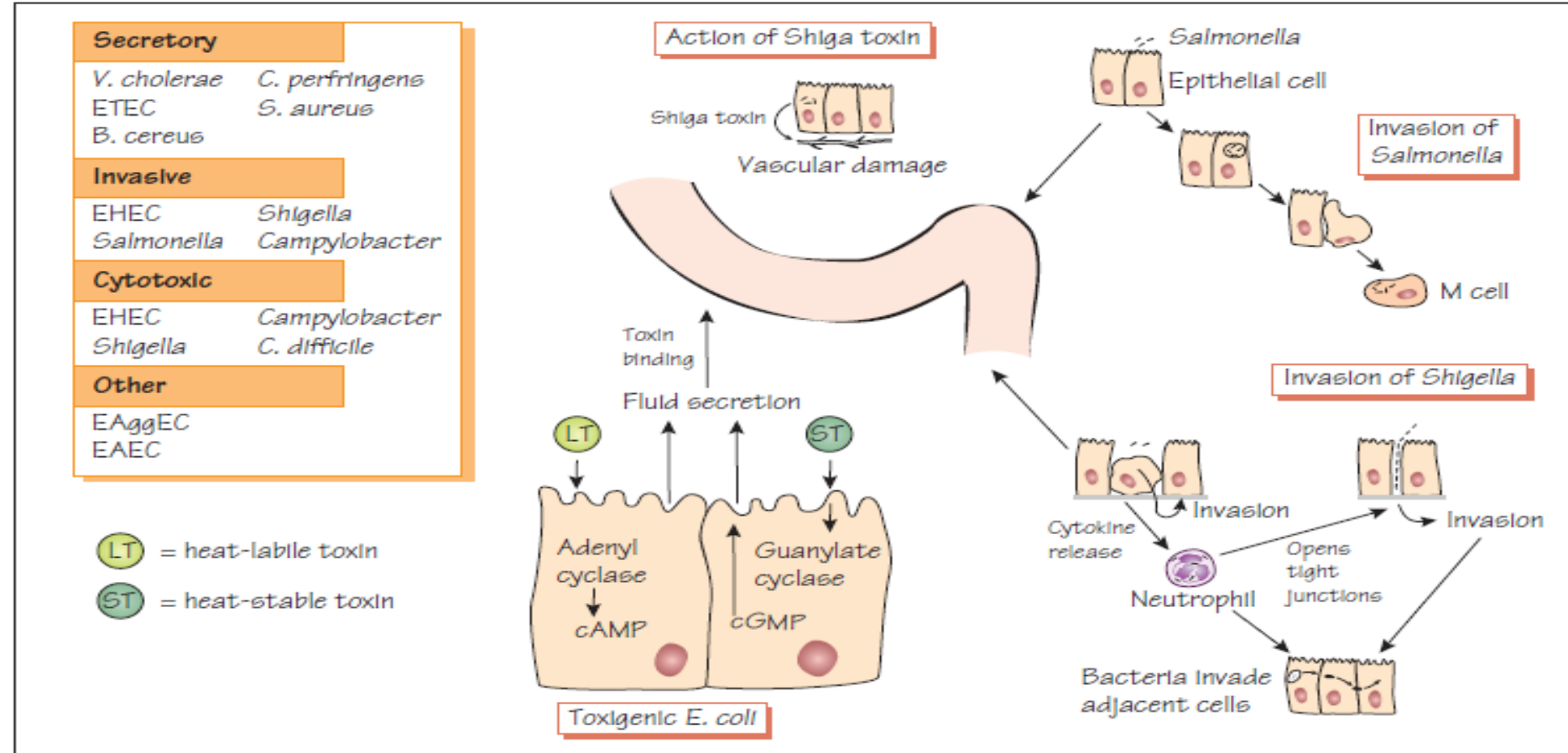
- ❑ Diarrhea is defined by the WHO as having 3 or more loose or liquid stools per day, or as having more stools than is normal for that person.
- ❑ Diarrhea is usually a symptom of an infection in the intestinal tract, which can be caused by a variety of bacterial, viral and parasitic organisms.
- ❑ Infectious diarrhea is common and a very important cause of morbidity and mortality in children under 5 years worldwide.
- ❑ Host defense mechanism
 - ✓ Gastric pH
 - ✓ Immunoglobulin A (IgA)
 - ✓ Host's normal gut biota
 - ✓ Peristalsis



Pathogenesis

- ❑ **Gastroenteritis can arise in two ways.**
 - **Direct infection** by consuming live pathogens. Some bacteria cause an infection following consumption of contaminated food.
 - **From food intoxication** via **ingestion of pre-formed bacterial enterotoxins** in contaminated food. The pathogen may secrete an exotoxin that contaminates the food and is then ingested by the host. This is referred to as a food intoxication because the toxin is ingested but the presence of living microorganisms is not required.
- ❑ **Toxin-mediated** dysregulation of intestinal cells causing fluid secretion.
- ❑ **Invasion** of the intestinal wall occurs with destruction of the cells.
- ❑ **Secretory diarrhoea** produces infrequent large-volume stools as the absorptive capacity of the colon is overwhelmed.
- ❑ In **dysenteric** illness (e.g. Shigella), colonic inflammation causes a loss in bowel capacity and frequent, often blood-stained stools.

Pathogenesis



Epidemiology

- ❑ Organisms are transmitted by **faecal–oral route**.
- ❑ Transmission is from poor hand hygiene or sanitation
- ❑ The **infective dose** can be as few as 10 organisms (*Shigella*).
- ❑ Some foods (e.g. milk) or drugs (e.g. H2 antagonists and proton pump inhibitors) may **reduce the protective effects of gastric acid**.
- ❑ Bacteria enter the food chain from infected animals, from **poor hygiene** during slaughter, and butchering. **Hens that are chronically colonized with *Salmonella*** may produce eggs that are contaminated.
- ❑ Transmission is favored by poor sanitation; infection can spread rapidly causing significant mortality.

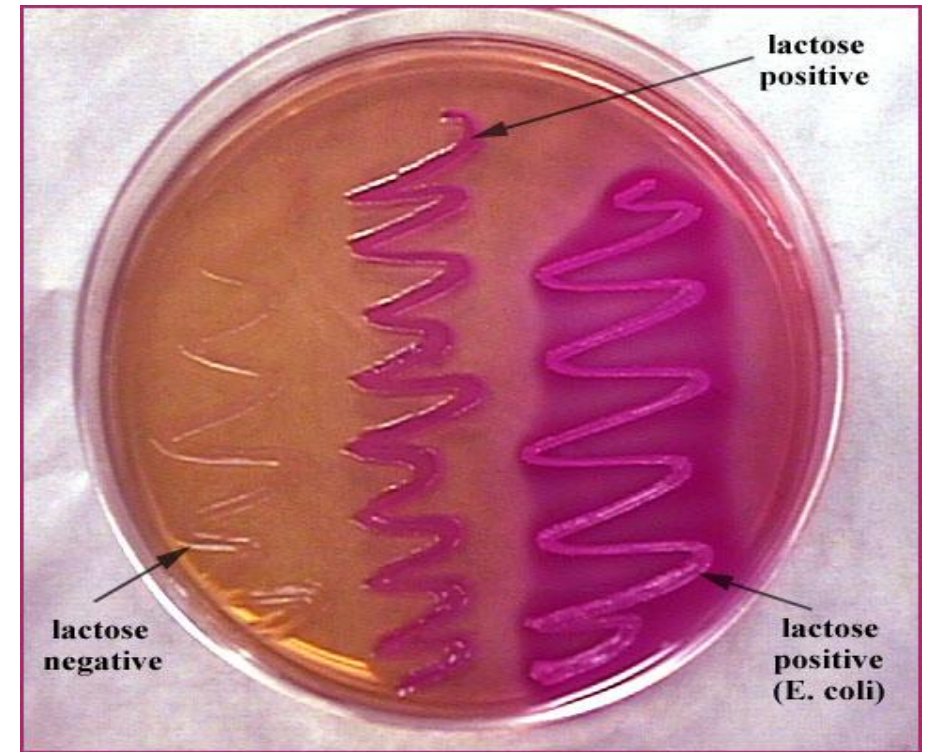
❑ The main genera inducing bacterial diarrhea are:

- *Escherchia*
- *Shigella*
- *Salmonalla*
- *Campylobacter*.

Escherichia coli

Important properties:

- ❑ Gram-negative bacilli (rods)
- ❑ Facultative anaerobes but grow best aerobically
- ❑ Most strains can **ferment lactose** (that is, they are Lac+). produces both acid and gas during fermentation of carbohydrates.

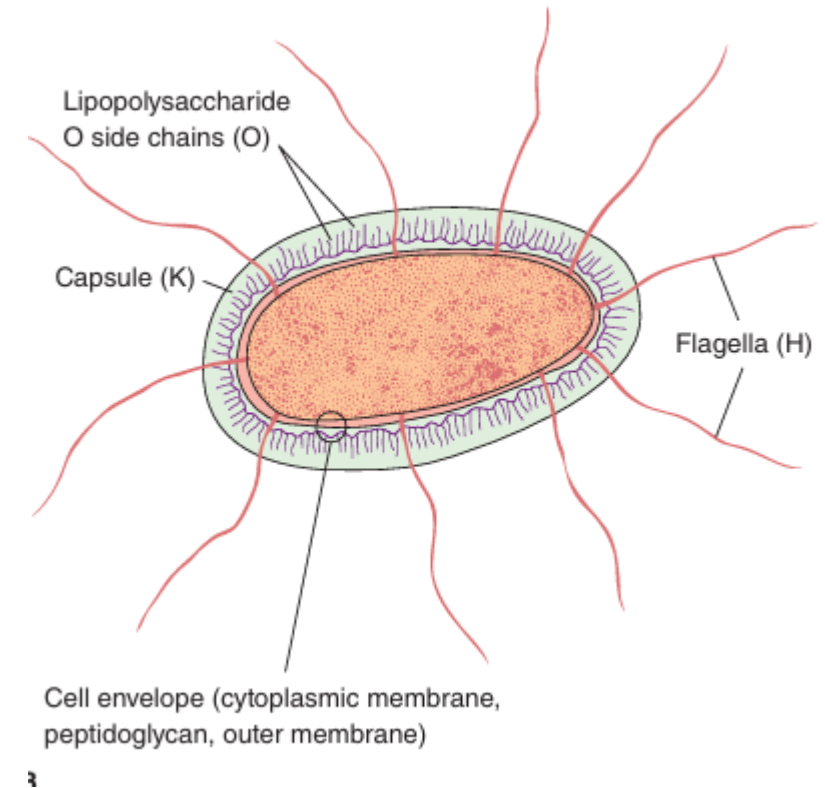


Lac+ and Lac -ve on MacConkey agar

- ❑ Member of the **Enterobacteriaceae family**, it is part of the normal flora of the colon in humans and other animals but **can be pathogenic both within and outside of the GI tract.**

***Escherichia coli* Gastroenteritis**

- ❑ A major causative agent of food-borne diarrhea.
- ❑ The vast majority of *E. coli* strains are harmless. Certain strains have become pathogenic by acquiring genes through horizontal gene transfer.
- ❑ These strains are readily tracked by serotyping. Two of the most commonly used serotyping antigens are the H (flagellar) and the O (O-antigen portion of LPS) antigens.



Antigenic structure of Enterobacteriaceae.

Pathogenesis

- ❑ The reservoir of *E. coli* includes both humans and animals.
- ❑ the *E. coli* that causes traveler's diarrhea is acquired by ingestion of food or water contaminated with human feces. Note that the main reservoir of enterohemorrhagic *E. coli* O157 is cattle and the organism is acquired in undercooked beef, for example, hamburgers.
- ❑ *E. coli* secrete exotoxins, called enterotoxins, that activate adenylate cyclase within the cells of the small intestine, causing diarrhea.
- ❑ The differences in the degree of virulence of various *E. coli* strains is correlated with the acquisition of plasmids, integrated prophages, and pathogenicity islands

- ❑ *Escherichia coli* has several clearly identified components that contribute to its ability to cause disease: pili, a capsule, endotoxin, and three exotoxins (enterotoxins).
- ❑ The first step is the adherence of the organism to the cells of the jejunum and ileum by means of pili.
- ❑ Once attached, the bacteria synthesize enterotoxins, which act on the cells of the jejunum and ileum to cause diarrhea.
- ❑ The toxins are strikingly cell-specific; the cells of the colon are not susceptible, probably because they lack receptors for the toxin.

E. coli associated diarrheal diseases

E. coli that cause diarrhea are extremely common worldwide, classified by the characteristics of their virulence properties, and each group causes disease by a different mechanism

1. Enteropathogenic *E. coli* (EPEC)

- They are an important cause of **diarrhea in infants**, especially in developing countries
- characterized by severe, **watery diarrhea**, vomiting, and fever, which are usually self-limited but can be prolonged or chronic.

2. Enterotoxigenic *E. coli* (ETEC)

- The most common cause of **travelers' diarrhea** and a very important cause of diarrhea in **children less than 5 years** of age in developing countries.
- Strains produce heat-stable enterotoxin that increases host cGMP concentrations, and heat-labile enterotoxin that has a structure and activity **very similar to cholera toxin**.
- Both toxins induce **hypersecretion of electrolytes** and water into the intestinal lumen, triggering **watery diarrhea**.



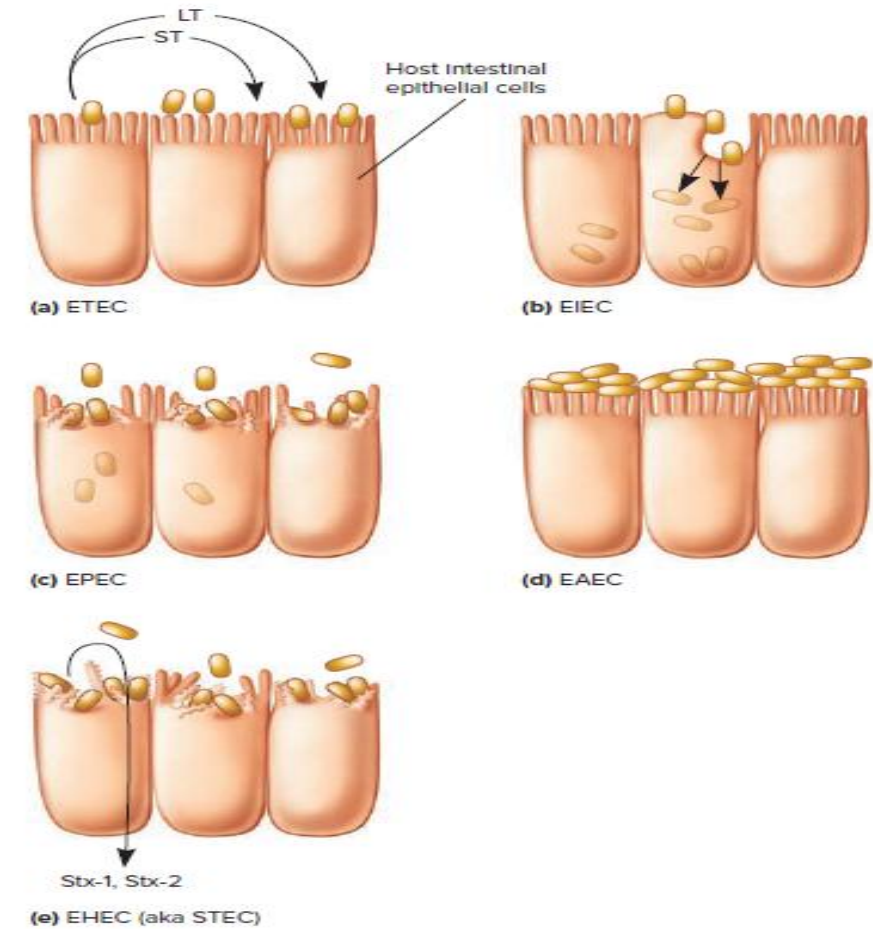
3. Shiga toxin-producing *E. coli* (STEC)

- Are named for the **cytotoxic toxins they produce**.
- STEC has been associated with **mild non-bloody diarrhea**, hemorrhagic colitis, and with **hemolytic uremic syndrome** (a disease resulting in acute renal failure).
- A **low infectious dose** (< 200 CFU) is associated with infection.
- Of the more than 150 *E. coli* serotypes that produce Shiga toxin, **O157:H7 is the most common** and is the one that can be identified most readily in clinical specimens.

4. Enteroinvasive *E. coli* (EIEC)

- Produce a disease very **similar to shigellosis**.
- The disease occurs **most commonly in children** in developing countries and in travelers to these countries.
- Similar to *Shigella*, EIEC strains are **non-lactose or late lactose fermenters** and are nonmotile.
- EIEC produce disease by **invading** intestinal mucosal epithelial cells.

5. Enteroaggregative *E. coli* (EAEC) causes acute and **chronic diarrhea** (>14 days in duration) in persons in developing countries. These organisms have been associated with traveler's diarrhea and **persistent diarrhea in patients with HIV**. They are characterized by their specific patterns of adherence to human cells.



Five Strains of Diarrheagenic *E. coli*. Each strain of diarrhea-causing *E. coli* can be classified by the nature of its interaction with host intestinal epithelial cells.

Laboratory Diagnosis

- Diagnosis of diarrhea caused by *E. coli* is based on travel history and symptoms.
- Laboratory diagnosis is by culture of the specific type of *E. coli* from feces and immunoassays to detect Shiga toxins.
- Specimens should be processed within 2 h of collection or stored at 4°C until processed.
- A Gram stain of fecal specimens is not useful? since the Gram-negative bacilli cannot be distinguished from the normal Gram-negative intestinal microbiota.

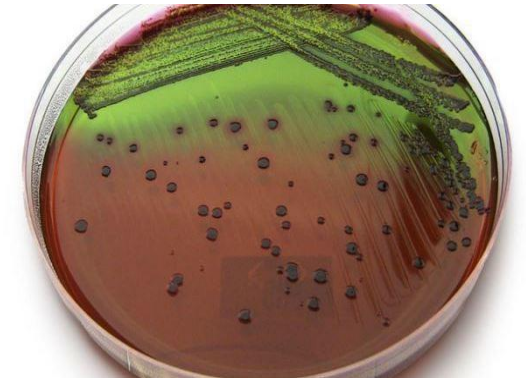


Rod shaped E.coli cells in Gram stain

- Specimens suspected of containing enteric gram-negative rods are grown on a differential medium, such as **EMB** agar or **MacConkey's** agar.
- *Escherichia coli*, which ferments lactose, forms **pink colonies**, whereas lactose-negative organisms are **colorless**. On EMB agar, *E. coli* colonies have a characteristic **green sheen**.
- Sorbitol MacConkey's agar contains sorbitol **instead of lactose**. *E. coli* 0157 produces **color less colonies** on this medium because it does not ferment sorbitol which serves as an important criterion that distinguishes it from other strains of *E. coli*.



E. Coli on MacConkey agar



E. Coli on EMB agar



Left, *Escherichia coli* 0157:H7 growing on MacConkey agar. Right, *E. coli* 0157:H7 on sorbitol MacConkey agar. *E. coli* 0157:H7 does not ferment sorbitol, whereas most other *E. coli* serotypes do ferment sorbitol.

Sorbitol MacConkey's agar

Treatment

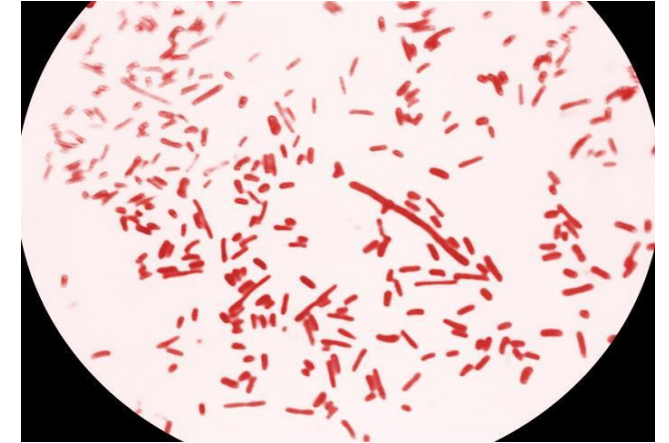
- Antibiotic therapy is usually *not* indicated in *E. coli* diarrheal diseases. However, administration of trimethoprim sulfamethoxazole or loperamide (Imodium) may shorten the duration of symptoms.
- Rehydration is typically all that is necessary in this self-limited disease.

Shigella species

Disease: Enterocolitis (bacillary **dysentery**). The term *dysentery* refers to bloody diarrhea.

Important properties

- Gram-negative rods
- Facultative anaerobes but grow best aerobically
- With the exception of ***Shigella sonnei***, they do **not ferment lactose**. The inability of lactose fermentation distinguishes shigellae on differential media from other, lactose-fermenting members of the Enterobacteriaceae (eg, *E. coli*).
- Distinguished from salmonellae by three criteria:
 - **produce no gas** from the fermentation of glucose
 - **not produce H₂S**
 - **nonmotile**.



Gram-negative rods (Bacilli)



Antigenic Structure

- Shigella have a complex antigenic pattern. There is great overlapping in the serologic behavior of different species, and **most of them share O antigens** with other enteric bacilli.
- The **somatic O antigens** of shigellae are **lipopolysaccharides**. Their serologic specificity depends on the polysaccharide. The classification of shigellae relies on **biochemical and antigenic** characteristics.
- The pathogenic species are ***S. sonnei*, *S. flexneri*, *S. dysenteriae*, and *S. boydii***

Pathogenesis

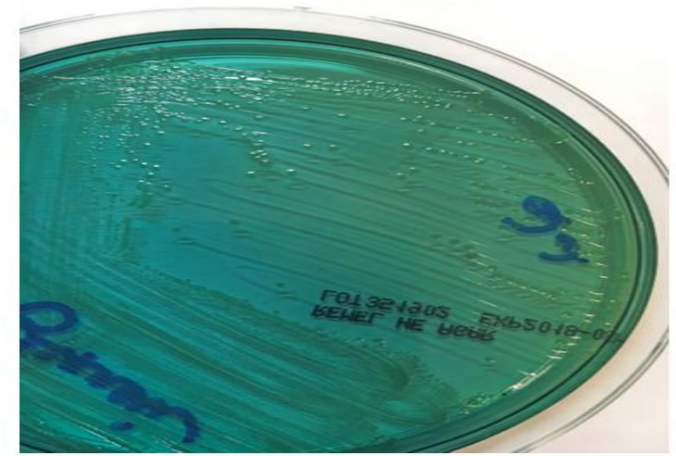
- Shigella infections are almost always limited to the gastrointestinal tract; bloodstream **invasion is quite rare**.
- Shigella are **highly communicable**; the infective dose is low, on the order of 10^2 organisms (in comparison, the infectious dose for salmonellae and vibrios is usually 10^5 – 10^8).
- Invasion of the mucosal epithelial cells (eg, M cells) by induced phagocytosis, escape from the phagocytic vacuole, multiplication and spread within the epithelial cell cytoplasm, and passage to adjacent cells.
- Microabscesses in the wall of the large intestine and terminal ileum lead to necrosis of the mucous membrane, superficial ulceration, bleeding, and formation of a “pseudomembrane”.
- .As the process subsides, granulation tissue fills the ulcers, and scar tissue forms

Clinical Finding

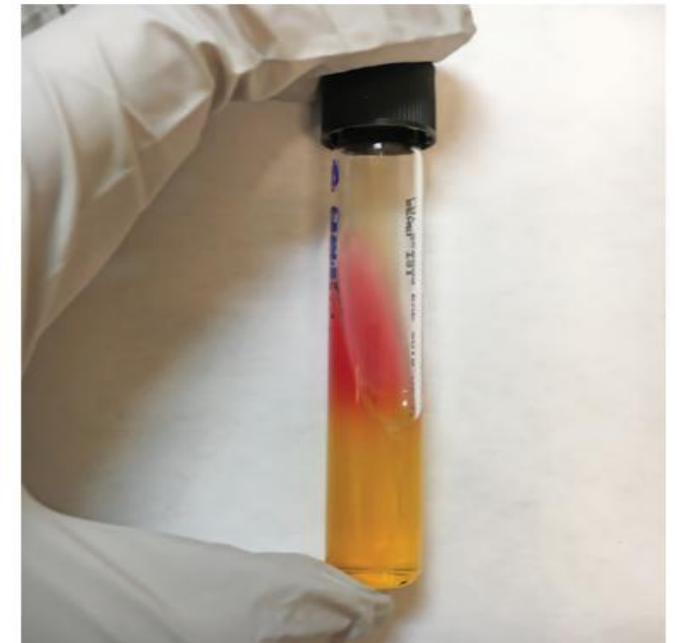
- After an incubation period of 1 to 4 days, symptoms begin with fever and abdominal cramps, followed by diarrhea, which may be watery at first but later contains blood and mucus.
- The disease varies from mild to severe depending on two major factors: the species of *Shigella* and the age of the patient, with young children and elderly people being the most severely affected.
- The diarrhea frequently resolves in 2 or 3 days; in severe cases, antibiotics can shorten the course.

Laboratory Diagnosis

- Form non–lactose-fermenting colonies on MacConkey's or EMB agar, and on **selective media**, such as **Hektoen Enteric agar**
- On **Triple Sugar Iron Agar (TSI)**, they cause an alkaline slant and an acid butt, with no gas and no H₂S.
- Confirmation of its group are done by slide agglutination
-
- A methylene blue stain of a fecal sample helps identify **neutrophils**, indicating an **invasive infection** (e.g., *Shigella*, *Salmonella*, *Campylobacter*) rather than a **toxin-producing organism** such as *V. cholerae*, *E. coli*, or *Clostridium perfringens*.



Shigella species on Hektoen Enteric (HE) agar



Shigella species on TSI agar slant

Treatment

- Shigellosis is a self-limited illness, and many patients recover without treatment within 5–7 days.
- Oral fluid replacement is considered to be sufficient for treatment of uncomplicated shigellosis, but in high-risk patient populations intravenous fluid replacement may be required.
- Antidiarrheal medications should be avoided in *Shigella* dysentery, as such medications may worsen the symptoms of the illness.
- Antibiotic treatment is recommended for the treatment of severe infections.

Salmonella species

Disease

- Cause **enterocolitis**, enteric fevers such as typhoid fever, and septicemia. They are one of the most common causes of **bacterial enterocolitis** in the United States.

Important properties

- Gram-negative bacilli that vary in length.
- Most isolates are motile .
- Facultative anaerobic,
- Not ferment lactose
- produce H₂S.

***Salmonella* species**

- Clinically, the *Salmonella* species are often thought of in two distinct categories:
 - Typhoidal species (i.e., those that cause typhoid fever) and
 - Nontyphoidal species (i.e., those that cause diarrhea [enterocolitis] and metastatic infection.
- *Salmonella typhimurium* is the most common species of *Salmonella* to cause **enterocolitis** in the United States, but almost every species has been involved.

Pathogenesis and Clinical Finding

- Acquired by the oral route. They are typically transmitted via contaminated water or food from animals and animal products to humans.
- The dose of Salmonella required is high, at least 100,000 organisms
- Enterocolitis is characterized by an invasion of the epithelial and subepithelial tissue of the small and large intestines.
- Neutrophils limit the infection to the gut and the adjacent mesenteric lymph nodes; bacteremia is infrequent in enterocolitis.
- After an incubation period of 12 to 48 hours, enterocolitis begins with nausea and vomiting and then progresses to abdominal pain and diarrhea.
- Usually the disease lasts a few days, is self-limited, causes nonbloody diarrhea.

Laboratory Diagnosis

- In enterocolitis, the organism is most easily isolated from a stool sample. However, in the enteric fevers, a blood culture is the procedure most likely to reveal the organism during the first 2 weeks of illness.
- No lactose fermenting colonies on MacConkey's or EMB agar.
- Produces H_2S , and the ferric ammonium citrate in the Hektoen enteric (HE) agar result in the Salmonella colonies to appear black.
- On TSI agar, an alkaline slant and an acid butt, frequently with both gas and H_2S (black color in the butt), are produced.



Salmonella species on HE agar



Salmonella on Triple Sugar Iron Agar

Treatment

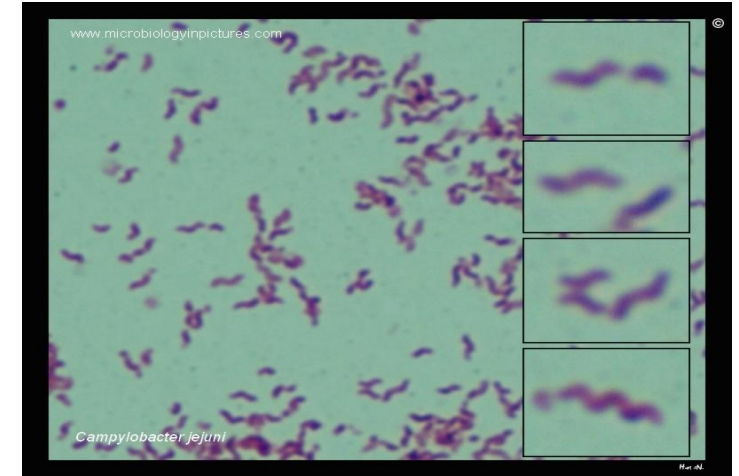
- Enterocolitis caused by *Salmonella* is usually a self-limited disease that resolves without treatment.
- Fluid and electrolyte replacement may be required.
- Antibiotic treatment does not shorten the illness or reduce the symptoms; in fact, it may prolong excretion of the organisms, increase the frequency of the carrier state, and select mutants resistant to the antibiotic.

Antimicrobial agents are indicated only for neonates or persons with chronic diseases who are at risk for septicemia.

Campylobacter jejuni

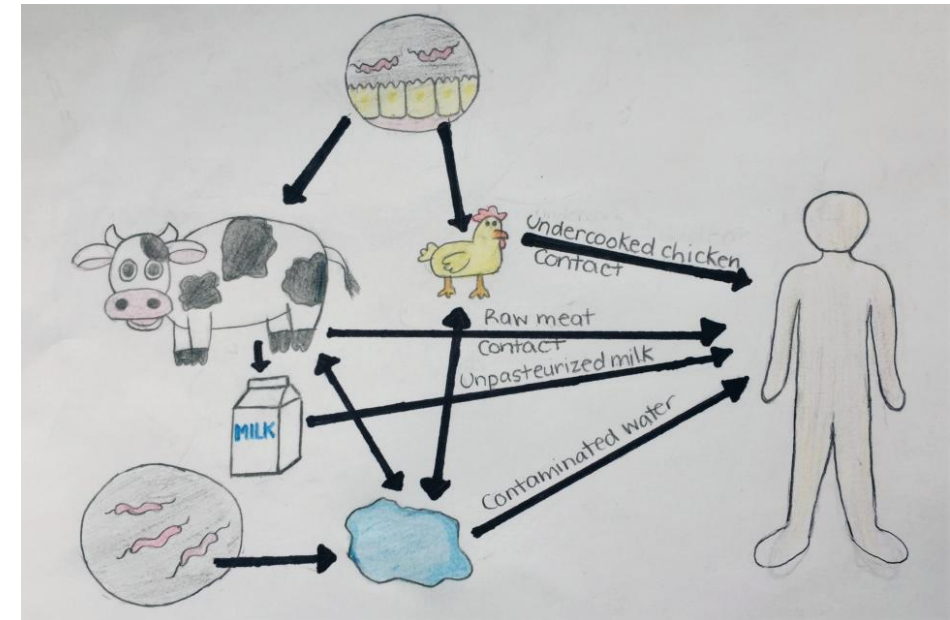
Important Properties

- Curved, gram-negative rods that appear either **comma-** or **S-shaped**.
- **Microaerophilic**, growing best in 5% oxygen rather than in the 20% present in the atmosphere.
- *Campylobacter jejuni* grows well at **42°C**, whereas *Campylobacter intestinalis* does not (an observation that is useful in microbiologic diagnosis).



Pathogenesis

- ❑ **Domestic animals** such as cattle, chickens, and dogs serve as a source of the organisms for humans. Transmission is usually **fecal–oral**.
- ❑ Foods, such as poultry, meat, and unpasteurized milk, are commonly involved. Puppies with diarrhea are a common source for children.
- ❑ Human-to-human transmission occurs but is less frequent than animal-to-human transmission.
- ❑ *Campylobacter jejuni* is a major cause of diarrhea in the United States; In U.S., *Campylobacter* enteritis > (*Salmonella* plus *Shigella*).
- ❑ **Low infectious dose (as few as 500)**
- ❑ **Invades** mucosa of the colon, destroying mucosal surfaces; blood and **pus** in stools (inflammatory diarrhea)



Clinical Finding

- ❑ The incubation period is **2 to 10 days**.
- ❑ **Enterocolitis**, caused primarily by *C. jejuni*, begins as watery, foul-smelling diarrhea followed by bloody stools accompanied by fever and severe abdominal pain.
- Recovery usually takes **5 to 8 days**.
- ❑ Gastrointestinal infection with *C. jejuni* is associated with **Guillain-Barré syndrome** in rare cases, in which the body's immune system attacks peripheral nerves, resulting in **life-threatening paralysis**.
- ❑ **The structural resemblance between *C. jejuni* surface polysaccharides and a sugar moiety found on nerve cell glycoproteins** has been linked to the development of Guillain-Barré syndrome in rare cases following *C. jejuni* infection through a mechanism termed **molecular mimicry**.

Laboratory Diagnosis

- ❑ Typical microscopic morphology (not easily visualized with the safranin; therefore, carbol fuchsin can be used as the counterstain, or extending the staining time of the safranin to at least 10 minutes).
- ❑ A stool specimen is cultured on Campylobacter Selective Agar (Skirrow) at 42°C, in a microaerophilic atmosphere containing 5% oxygen and 10% carbon dioxide.
- ❑ It is identified by failure to grow at 25°C, **oxidase positivity**, and **sensitivity to nalidixic acid**.



Skirrow agar

Treatment

- Mostly supportive via fluid and electrolyte replacement
- Erythromycin, fluoroquinolones, penicillin resistant

Prevention

- Proper food preparation (particularly poultry),
- Avoidance of unpasteurized dairy products
- Implementation of safeguards to prevent contamination of water supplies.

Case study

A group of teenagers became ill with nausea, vomiting, severe abdominal cramps, and diarrhea after eating undercooked hamburgers from a local restaurant. Two of the teenagers were hospitalized with hemolytic-uremic syndrome. *E. coli* O157:H7 was isolated from the patient's stools as well as from uncooked hamburgers. The O157 refers to which bacterial structure?

- A. Lipopolysaccharide
- B. Capsule
- C. Flagella
- D. Fimbriae
- E. S-layer

A woman has been in the hospital for 3 weeks due to complications following surgery for colon cancer. Five days into a course of ceftriaxone for suspected pneumonia she developed diarrhea. Colonoscopy revealed multiple plaques on the mucosa which are composed of fibrin and WBS. Stool examinations for *Salmonella*, *Shigella*, *Campylobacter*, and amoebas are negative. The diarrhea and plaques are most likely produced by:

- A. Endotoxin from clostridial spores
- B. Exotoxin from clostridial spores
- C. Exotoxin from clostridial cells
- D. Exotoxin from *E. coli* cells
- E. Endotoxin from *E. coli* cells

Which one of the following host defense mechanisms is the MOST important for preventing dysentery caused by Salmonella?

- A. Gastric acid
- B. Salivary enzymes
- C. Normal flora of the mouth
- D. Alpha interferon
- E. Complement system

The pathogenesis of which one of the following organisms is MOST likely to involve invasion of the intestinal mucosa?

- (A) *Vibrio cholerae*
- (B) *Shigella sonnei*
- (C) *Enterotoxigenic Escherichia coli*
- (D) *Clostridium botulinum*
- (E) *S. aureus*

In salmonella food poisoning:

- A.** Symptoms start within 3 hours
- B.** Antibiotic treatment is not indicated
- C.** Infected food-handlers are the main source of contamination
- D.** Diagnosis is usually confirmed by culture of the suspected food.
- E.** Blood cultures are frequently positive

A - F Usually takes 24-48 hours. The shorter incubation period is associated with exotoxin-producing organisms such as *Staphylococcus aureus* and *Bacillus cereus*

B - T Self-limiting in most cases

C - F Source is often contaminated food such as poultry, eggs, etc

D - F Usually confirmed by culture of the diarrhoeal stool

E - F Not often positive in food poisoning

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